W–L–200

NOVEMBER 18, 1955

SUPERSEDING
Int. Fed. Spec. W–L–00200 (GSA–FSS)
January 14, 1955

FEDERAL SPECIFICATION

LANTERN, ELECTRIC

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope.—This specification covers electric hand lanterns of the dry-cell battery operated type. Waterproof and explosion-proof lanterns and batteries for lanterns are not covered by this specification.

1.2 Classification.

1.2.1 *Types.*—The electric lanterns shall be of two types as follows:

Type I.—Single beam. (Concentrated light beam.)

Type II.—Two beam. (One concentrated light beam and one diffused light beam.)

2. APPLICABLE SPECIFICATIONS AND STANDARDS

2.1 The following specification and standard of the issues in effect on date of invitation for bids, contract or order, form a part of this specification:

Federal Specification:

W–B–101—Batteries and Cells; Dry. (See 3.5.)

Military Standard:

MIL–STD–105—Sampling Procedures and Tables for Inspection by Attributes. (See Sec. 4.)

(Activities outside the Federal Government may obtain copies of Federal Specifications and Standards as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, Washington, D. C.

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(Copies of Military Standard required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Preproduction sample.—When so specified in the invitation for bids, a preproduction sample of the lantern shall be submitted to the contracting officer for test and approval (see 6.2).

3.2 Material.—All materials shall be suitable for the use intended and shall be free from defects which might affect their serviceability or appearance.

3.3 Construction.—The lanterns shall be of sturdy construction capable of meeting the test requirements of section 4.

3.4 Finish.—All external metal parts of the lantern shall be corrosion-resisting or have a corrosion-resisting finish applied by plating or enameling.

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3.5 Design, general.—The lanterns shall be complete units including battery container, light source or sources, switch and handle. They shall utilize the current drawn from one 6-volt lantern battery (battery not included with lantern) designated 4F in Federal Specification W-B-101. The lantern shall be capable of being carried by the handle, or of standing on the base without other support. The concentrated light source shall be capable of projecting its beam horizontally without interference from the lantern either when standing or being carried. The switch shall provide a positive instantaneous means of energizing or de-energizing the light source.

3.5.1 *Type I.*—The type I lantern shall be capable of producing and projecting a concentrated light beam of nearly parallel rays having characteristics which conform to 3.8.1.

3.5.2 *Type II.*—The type II lantern shall be capable of producing and projecting a concentrated light beam equal to that of type I lanterns (see 3.5.1) and also an independent diffused light beam having characteristics which conform to those required in 3.8.2. The switch shall provide a positive instantaneous choice of either light source.

3.6 Component parts.

3.6.1 *Battery container.*—The battery container shall be of metal and of a size and shape to accommodate one 4F 6-volt lantern battery (see Federal Specification W-B-101) and other necessary wiring and contacts. The container lid shall be suitably fitted to retain the battery in proper position when subjected to the tests in section 4. No tools shall be required to remove or replace the battery.

3.6.1.1 *Instructions.*—The battery container shall have suitable instructions permanently affixed inside for proper installation of battery. The instructions shall be embossed, stenciled, or printed.

3.6.2 *Handle.*—The lantern shall have a suitable handle of metal for carrying the lantern in the hand. The handle shall distribute the weight of the lantern without excessive strain on any part and shall not interfere with the projected light beam or beams when the lantern is standing on its base.

3.6.3 *Switch, general.*—There shall be a switch provided in the electrical circuit which shall be completely enclosed except for its actuating member. The actuating member shall be on the outside of the lantern and the type II lantern respective switch positions shall be suitably and permanently marked on the exterior of the lantern. If of metal, the exposed actuating member shall be suitably insulated from the electrical circuit to prevent accidental discharge of the battery through contact with any exterior metal part of the lantern. The switch shall be constructed to resist accidental movement from one position to another. If switch blades close the lantern circuit by making contact with internal metal parts, those metal parts shall be insulated from contact with any exterior metal part of the lantern.

3.6.3.1 *Switch, type I.*—The switch provided in type I lanterns shall have an "on" and an "off" position. In addition, a flash or signaling switch position may be provided.

3.6.3.2 *Switch, type II.*—The type II lantern switch shall have not less than three positions including "on", "off", and "on", to permit operation of either the concentrated or diffused light beam.

3.6.4 *Reflector head assembly, general.*—The reflector head assembly shall consist of those parts of the lantern which accomplish conversion of the electrical energy of the battery into visible light and the projection of that light as a concentrated, or as a diffused beam of light. The essential parts of the assembly shall be a lamp, lampholder, reflector, lens, reflector and lens retaining ring, and assembly casing. All ferrous-metal parts of the reflector head assembly except current-carrying parts shall have a baked-enamel or electro-plated finish to prevent corrosion.

3.6.4.1 *Concentrated light, reflector head assembly, type I.*—Type I lanterns shall be equipped with prefocused light sources either of the sealed beam type with integral reflector, lens, filament and screw-type terminals, trade No. 4546, PAR-36, 5 volts, 0.50 ampere or trade No.

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4506, PAR–36, 4.7 volts, 0.47 ampere or approved equal; or have separate reflectors which are essentially parabolic in shape and have an aluminized or silver plated specular finish. Silver-plated reflectors shall be protected by clear lacquer. The separate reflector shall be used with a single contact, miniature, flanged base lamp trade No. PR–15, 4.82 volts, 0.50 ampere or trade No. PR–13, 4.75 volts, 0.50 ampere, or approved equal, and a lampholder so located with respect to the reflector that when the lamp and reflector are in fully assembled position, the lamp filament is approximately at the focal point of the reflector. The lampholder shall hold the lamp in position by means of a suitable cap and the electrical contact with the lamp base shall be made by a helical coiled spring or other suitable means to prevent transmission of physical shock to the lamp base. The lens, when separate, shall be of clear glass; or plastic, if so specified (see 6.2) not less than 0.117 inch thick and be free of bubbles, striae, checks or cracks and the effective lens area for light transmission with the retaining ring in place shall be not less than 2¾ inches in diameter.

3.6.4.1.1 *Reflector head assembly movement, type I.*—The reflector head assembly of type I lanterns shall be capable of being rotated through a vertical angle of not less than 90° when the lantern is resting on the base. The angle through which the reflector head assembly rotates shall be measured upward from 10° below the horizontal when the lantern is resting on its base on a horizontal plane. The reflector head assembly shall remain as positioned when the lantern is subjected to movements similar to those expected during normal use.

3.6.4.2 *Diffused light reflector head assembly, type II.*—The diffused light beam producing reflector head assembly shall be provided with a miniature screw-base lamp, trade No. 502, 5.1 volts, 0.15 amperes or approved equal. The lampholder shall be a miniature screw base shell which may be an integral part of the reflector. The reflector shape shall be at the option of the manufacturer and the reflector surface shall be silver plated or aluminized with a matte or semi-matte finish if the lens provided is of clear glass or plastic. The reflective surface of the reflector shall be specular if the lens provided is of the bulls-eye or honeycomb type. The lens shall be of glass; or plastic, if so specified (see 6.2), not less than 0.117 inch thick. The effective light transmitting area of the lens shall be not less than 2¾ inches, nor more than 3½ inches in diameter. A suitable metal guard shall be provided to protect the reflector head assembly. The guard shall not obstruct the light beam and be sturdy enough to withstand the rigidity test given in 4.4.6. The diffused light reflector head assembly shall be capable of projecting its beam of light in a vertical direction when the lantern is standing on its base.

3.6.5 *Base.*—The lanterns shall have a suitable base which will provide a stable platform when subjected to test under the conditions stated in 4.4.5 and 4.4.6. The base is defined as that surface which is intended by the manufacturer to serve as the support or bottom of the lantern.

3.7 Wiring and connections.

3.7.1 *Current-carrying parts.*—Internal current-carrying parts shall be of brass or phosphor bronze, and shall be of suitable cross-sectional area for the purpose intended. Bare, fixed conductors shall be securely held in position by rivets or other suitable means and be suitably insulated from contact with any metallic part of the outside of the lantern. Eyelet rivets or other fasteners shall not be used on current carrying parts if they produce openings to the interior of the battery container or reflector head assembly.

3.7.2 *Lamp lead wires.*—Lamp lead wires, if used, shall be flexible, rubber or plastic insulated not lighter than No. 22 AWG., and the terminal ends shall be permanently attached to the lampholder terminals; or in the case of sealed beam lamps be provided with permanently attached closed wire terminal lugs of suitable size for connection to the lamp terminals. Lamp lead wires shall be of ample length to permit replacement of parts and to permit normal movement of lantern parts without binding, chafing, or rubbing.

3.7.3 *Circuits.*—The lantern electrical circuits shall be designed so that it shall be impossible to energize the lamp by making contact

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between any two points on the exposed exterior surface of the lantern when subjected to the test described in 4.4.8.

3.7.4 *Chafing protection.*—All openings in the battery container or reflector head assembly casing through which lead wires pass shall be smooth and well rounded or be provided with grommets or other suitable means to prevent chafing.

3.8 Performance.

3.8.1 *Light characteristics, concentrated light, type I and type II.*—The light beam produced by the concentrated light reflector head assembly shall consist of essentially parallel rays. When projected onto a screen whose plane is perpendicular to the optical axis of the light assembly being tested and 5 feet distant from the face of the lens the beam shall be not greater than 9 inches in diameter and be relatively free of shadows and filament images. Conformance of light characteristics shall be determined under the conditions stated in 4.4.2.

3.8.2 *Light characteristics, diffused light, type II.*—The type II lantern diffused beam of light shall, when tested under the conditions of 3.8.1 and 4.4.3, produce an image on the screen not less than 60 inches nor more than 80 inches in diameter. The image shall be essentially circular with no shadows from the guard or handle.

3.8.3 *Switch.*—The switch shall be capable of completing 25,000 cycles of operation without breakage or apparent wear or becoming electrically inoperative. A cycle shall be from "off" through all "on" positions and back to "off." (See 6.3.)

3.8.4 *Stability.*—The assembled lantern, including battery, shall be capable of standing without other support and shall not tip, fall or roll when placed on its base on a plane surface inclined 15° from the horizontal in the positions stated in 4.4.5.

3.8.5 *Rigidity.*—While standing on its base on a horizontal plane, the lantern shall be capable of withstanding a compressive load of 50 pounds in accordance with the requirements of 4.4.6.

3.9 Workmanship.—The lantern workmanship shall be free from defects which will affect the appearance of the lanterns or impair their serviceability.

4. SAMPLING, INSPECTION AND TEST PROCEDURES

4.1 Sampling for lot acceptance.

4.1.1 *Inspection lot.*—All lanterns of the same type presented for inspection at one time shall be considered as a lot for purposes of inspection and tests.

4.1.2 *Sampling for inspection.*—A representative random sample shall be taken from the lot in accordance with table I. Acceptance or rejection of the lot shall be based on the number of defective units found in the sample selected and inspected, as being equal to, or less than, the specified acceptance number; or, equal to, or greater than, the specified rejection number of table I.

TABLE I.—*Sampling for visual and dimensional inspection, and concentrated and diffused light beam tests. Based on Accepted Quality Level (AQL) of 2.5 per cent*

Lot size	Sample size	Acceptance number	Rejection number
15 and under ..	3	0	1
16 to 40	7	0	1
41 to 110	15	1	2
111 to 300	35	2	3
301 to 500	50	3	4
501 and over ..	75	4	5

NOTE—The Government inspector may institute tightened or reduced inspection in accordance with Military Standard MIL-STD-105.

4.1.3 *Sampling for tests.*—A representative random sample shall be taken from the lot in accordance with table II. Acceptance or rejection of the lot shall be based on the number of defective lanterns found in the sample selected and tested as being equal to or less than the

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specified acceptance number; or, equal to, or greater than, the specified rejection number of table II.

TABLE II.—*Sampling for tests*

Lot size	Sample size	Acceptance number	Rejection number
500 and under .	3	0	1
501 to 1300	4	0	1
1301 to 3200 ..	5	0	1
3201 to 8000 ..	7	0	1
8001 and over .	10	0	1

4.2 Inspection.—Each of the sample lanterns selected in accordance with table I shall be visually and dimensionally inspected by the Government inspector to verify compliance with this specification. (See 4.4.1.) Any lantern in the sample having one or more defects shall be rejected, and if the number of defective lanterns in any sample exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected.

4.3 Lot acceptance test by Government inspector.—Each of the sample lanterns selected in accordance with table II, except sample lanterns to be subjected to the concentrated light beam and diffused light beam tests (see 4.4.2 and 4.4.3) shall be selected in accordance with table I, shall be subjected to the following tests in any order which is satisfactory to the Government inspector. Any sample lantern which does not meet the requirements of all required tests shall cause rejection of the lot represented by the sample.

4.4 Test procedures.

4.4.1 *Visual and mechanical inspection.*—The lantern shall be inspected for workmanship, mechanical fit of parts, and miscellaneous defects. Switches, latching, and mechanical adjustments shall be inspected for workmanship. Clearances and dimensions shall be measured and checked against an approved contractor's drawing containing all dimensions and tolerances and markings checked.

4.4.2 *Concentrated light beam test.*—The lanterns shall be tested for compliance with the concentrated light performance requirements of 3.8.1. The test shall be conducted in a dark room and the lantern shall be equipped with fresh batteries.

4.4.3 *Diffused light beam test, type II.*—Compliance with the diffused light performance requirements of 3.8.2 shall be determined for type II lanterns in a dark room with fresh batteries installed.

4.4.4 *Switch test.*—The lanterns with batteries and lamps in place shall have each switch operated in a normal manner at a rate not to exceed 20 cycles per minute for a total of 1,000 cycles. A cycle shall be from "off" through all "on" positions and back to "off." Mechanical defects or faulty electrical operation of the switch shall constitute failure of the test. Lamps and batteries shall be replaced as required during the test. Failure of the lamps or batteries shall not be grounds for rejection.

4.4.5 *Stability test.*—The lanterns with batteries in place shall be placed on a plane inclined as stated in 3.8.4 and shall be free standing when placed in any of seven different equally spaced positions about an axis perpendicular to the plane of the inclined surface. The surface of the plane shall be rough enough to prevent sliding.

4.4.6 *Rigidity test.*—The assembled lanterns shall be subjected to an evenly distributed compressive load of 50 pounds applied vertically without shock to the top of the lantern. (Type II lanterns shall have the compressive load applied to the diffused light guard when the diffused reflector head assembly is pointing vertically.) The load shall remain for five minutes. Examination shall disclose no permanent deformation or breakage of parts as a result of the test.

4.4.7 *Drop test.*—The lantern with battery installed shall be dropped six times from a height of three feet into a box of loose, dry sand having a minimum depth of one foot and resting on a stable horizontal base. The lantern shall fall three times each on the base and top. No per-

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manent deformation, breakage, except lens, or operating difficulty shall occur as a result of this test. The lid of the battery case shall remain fastened during the test. The surface of the sand shall be leveled after each drop.

4.4.8 *Circuit test.*—The assembled lantern with fresh battery installed and the switch in the "off" position shall be placed in all possible positions on a flat steel surface not less than 12 by 12 inches in area. The lamp shall not be energized as a result of this test.

5. PREPARATION FOR DELIVERY

5.1 Packaging.—Each lantern shall be individually packaged. Unless otherwise specified, commercial packages will be acceptable.

5.2 Packing.—Unless otherwise specified, lanterns shall be prepared for shipment to permit acceptance by common or other carrier for transportation at the lowest applicable rate to afford adequate protection from normal hazards of transportation.

5.3 Marking.

5.3.1 *Issue packages.*—Unless otherwise specified, issue packages shall be marked to include the name, type of lantern and the name of the manufacturer.

5.3.2 *Shipping containers.*—Unless otherwise specified, shipping containers shall be marked with the name and type of lantern, the name of the contractor, the number of the contract or order, the name and address of the consignee and the gross weight.

6. NOTES

6.1 Intended use.

6.1.1 Type I lanterns are intended for use in the illumination of objects at a distance as in general outdoor use.

6.1.2 Type II lanterns can be used for distant illumination or for illumination over a relatively wide area of objects closer to the observer.

6.2 Ordering data.—Purchasers should exercise any desired option offered herein (see 5.2 and 5.3) and should specify the following:

(a) Type (see 1.2.1).

(b) Plastic lens, if required (see 3.6.4.1 and 3.6.4.2).

(c) Preproduction sample, if required (see 3.1).

6.3 Certification.—A certificate, signed by a responsible representative of the bidder, certifying that the lanterns to be furnished comply with the requirements of this specification may be required by the purchasing officer in the invitation for bids.

6.4 It is believed that this specification adequately describes the characteristics necessary to secure the desired material, and that normally no samples will be necessary prior to award to determine compliance with this specification. If, for any particular purpose, samples with bids are necessary, they should be specifically asked for in the invitation for bids and the particular purpose to be served by the bid sample should be definitely stated, the specification to apply in all other respects.

6.5 This specification does not include all types, classes, grades, sizes, etc., of the commodity indicated by the title of the specification, or which are commercially available, but is intended to cover the types, etc., which are normally procured for Federal Government requirements.

6.6 Transportation description. — Transportation description applicable to this item is:

Lanterns or flashlights, electric

Carload minimum weight 20,000 pounds subject to Rule 34, Uniform Freight Classification

Truckload minimum weight 24,000 pounds

Minimum weights shown apply when material is transported at classification ratings. They will vary when material moves on commodity rates. When in doubt, consult General Services Administration Regional Traffic Manager.

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Patent notice.—When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

MILITARY INTEREST:
Army—Sig

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charger cord, not less than 9 feet in length, and integral cigarette lighter